

Text as Data for Social Scientists

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Why use Text as Data?

Common Typology in Text as Data

Manual Text Classification

Fully Automated Text Analysis

Semi-Automatic Text Analysis

Summary & Coding Session

Why use Text as Data?

- Run the package installation in the R-File
- It takes a moment, so let us have it run in the background

The Advantages of Text-based Data - 1

- Able to computationally analyze millions of documents (Grimmer et al. 2022)
 - Previously unfeasible research questions become accessible
 - Low costs compared to crowd-coding of texts
- Replicability (Grimmer et al. 2022)
 - Most methods can be fully replicated and always yield the same findings
- Text as a Naturally Created Data Source (Salganik 2017)
 - Text is produced independently of the researcher
 - No social desirability bias or predetermined answers

The Advantages of Text-based Data - 2

- Reduction of Human Coder Bias (Lacy et al. 2015)
 - No fatigue, interpretation bias, or inter-coder disagreement once validated
 - Validated models can be applied to further data
- Discovery & Hypothesis Generation
 - Unsupervised methods surface unanticipated patterns → e.g. for theory-building

Common Typology in Text as Data

Common Units of Analysis - 1

- Characters & Token: individual letters or subwords (Grimmer et al. 2022)
 - Rarely used as the primary unit in social science
- Words & Terms: the most common unit → foundation of bag-of-words approach
- Sentences: common unit in sentiment analysis and stance detection
 - Paragraphs: used in argumentation and discourse analysis

Common Units of Analysis - 2

- Documents: the full text as a single observation (holistic approach) (Grimmer et al. 2022)
 - e.g., speeches, newspaper articles, manifestos, or tweets
- Corpus: the entire text collection treated as one unit
 - Used for cross-corpora comparisons (e.g., partisan vs. neutral press) (Boydston et al. 2014)
- **The choice of unit directly shapes which methods are applicable**

Common Text Transformations - 1

- Bag-of-Words: counts word frequencies but ignores word order (Grimmer et al. 2022)
 - Document Feature Matrix (DFM) → Table with words and their frequencies per text
 - Tradeoff between simplicity and loss of information
 - TF-IDF weights terms by frequency relative to their occurrence across documents
- N-grams: captures short sequences of adjacent words to preserve some order
 - e.g., bigrams (two-word-sequence) or trigrams (three-word-sequence) (Manning et al. 2008)

Common Units of Analysis - Example of DFM

	doc_id	well	jim	first	like	thank	sponsor	debat	peopl	boston	host	governor
1	1	12	14	15	19	3	1	2	27	1	1	37
2	2	13	11	9	17	0	0	0	20	0	0	15
3	3	7	7	7	15	3	0	5	20	0	0	29
4	4	38	18	16	32	4	0	0	24	0	2	0
5	5	18	0	11	15	4	0	1	23	0	0	0
6	6	21	1	9	10	4	0	5	36	0	2	0
7	7	27	5	8	14	5	0	3	27	0	0	50
8	8	16	0	9	15	0	0	0	25	0	0	62
9	9	15	0	12	9	2	1	3	19	0	0	39
10	10	26	0	10	22	1	0	2	37	0	0	0
11	11	20	0	6	30	0	0	1	49	0	0	0
12	12	28	0	9	10	6	0	1	53	0	0	0
13	13	29	0	5	24	3	0	1	68	0	0	8
14	14	19	0	10	22	1	0	0	49	0	0	3
15	15	25	3	8	9	4	1	1	30	0	0	10
16	16	16	4	5	25	5	1	4	60	0	0	2
17	17	20	1	7	18	7	1	2	64	0	0	6
18	18	25	4	6	10	5	0	2	31	0	1	0
19	19	24	0	9	15	14	2	1	26	0	0	0
20	20	27	0	8	6	6	0	3	39	0	0	3
21	21	42	0	8	5	2	0	4	33	0	1	0

- DFM based on presidential debates data by Martherus 2020

- Common pre-processing pipeline (Denny and Spirling 2018)
 - Tokenisation
 - Lower-casing
 - Stopword removal
 - Common words with no meaning
 - Stemming
 - Reduce word to base, e.g., “stemming”, “stems”, and “stemmed” all become “stem”
- Pre-processing choices must be reported

Common Levels of Supervision

- Unsupervised: No pre-defined labels, algorithm finds structure (Blei et al. 2003)
 - E.g., topic models or clustering
 - Useful for exploration
- Semi-supervised: Small hand-coded datasets used to guide the algorithm (Grimmer et al. 2022)
 - Reduces labelling cost while keeping the human component
- Supervised: Fully human-labelled to train a machine learning model (Grimmer et al. 2022)
 - Validation against another human-labelled dataset (precision, recall, F1)
- All methods should be validated against human coding (Grimmer et al. 2022)

Common Types of Analyzed Texts

- Political texts: Speeches, party manifestos, parliamentary debates, press releases (Benoit and Laver 2003)
 - Large, naturally created, and digitized data sources
- News & Media: Newspaper articles, TV transcripts, online blogs (Boydston et al. 2014)
 - E.g., agenda-setting, framing, or media bias research
- Social Media: Twitter/X posts, Reddit threads, Facebook comments (Salganik 2017)
 - Careful: often comes with sample bias
- Open-ended Survey Questions & Historical Documents

Manual Text Classification

Advantages and Disadvantages

- Advantages of manual classification (Krippendorff 2019)
 - Gold standard for validity
 - Human recognizes nuance, context, irony, and implicit meaning
 - Flexible to emerging categories
 - Useful for theory building
 - Researcher can adapt coding rules when edge cases arise
 - Required for validation dataset
- Disadvantages of manual classification (Lacy et al. 2015)
 - Too costly (time and money) to apply to large text collections
 - Prone to coder fatigue and inter-coder disagreement over time
 - Hard to replicate reliably
 - Unconscious researcher bias possible
 - Possible language barriers

Close Reading & Content Analysis

- Close Reading: careful, interpretive reading to identify themes, meaning, and context
 - No standard procedure: validity rests on the researcher's interpretive skill
- Qualitative Content Analysis (Mayring 2022)
 - Systematic, rule-guided reduction of text to categories → Coding scheme required
- Discourse Analysis (Wodak and Meyer 2001)
 - Examines how language constructs social reality and power
 - Focuses on what is said, but also what is left unsaid
 - Language reflects but also forms social reality
 - Extensions:
 - Critical Discourse Analysis (CDA) (Mair and Fairclough 1997)

Qualitative Analysis in MAXQDA

The screenshot displays the MAXQDA software interface. The top bar shows the document name 'Riley (30 Paragraphs)' and a search bar. Below the top bar, the 'Work situation' section is active, showing a list of paragraphs. The main text area displays the content of paragraph 3, which is an interview transcript. The transcript includes a question (Q) and a response (R) from Riley. The response describes Riley's typical workday, mentioning flexibility, meetings, and a project. The right sidebar shows a list of segments, including 'Interviews > Berlin > Mateo' and 'Work situation > Flexibility (0)'. The bottom of the sidebar shows a list of segments, including 'Mateo_Pos_3' and 'Work situation > Flexibility (0)'.

Interview Riley

Q: Thank you very much for your participation! First, I would like to ask you to describe a typical workday. What does a typical workday look like for you?

R: Well, first of all it starts with the fact that I don't always come to the office at a fixed time, because I have the possibility to do flextime. That's why it's always a bit different. First of all, and most importantly: I get a coffee. Of course. Then I read through the e-mails and messages that arrived, answer them, that is probably like everywhere else. I also chat with my team. What else happened yesterday? Or is there any new information? What's up today? Exactly. Apart from that, it all depends on the phase we are at. But mainly sitting at the computer, reading a lot and sometimes there are meetings, but rather in small groups. It's not like in a big company where you would have huge meetings, but rather that in a group of three or four you get together if you want to do a project together. And then at some point it's time to go home.

Q: How long do you work? When do you finish work?

R: After eight hours. But of course, it often happens that you haven't finished reading a few things or something, which has to be done on the return trip in the train or when there is an important deadline than I usually have some work left to finish it at home.

Q: And what do you do in your free time after work?

34 coded segments (from 5 documents, 1 document group)

Interviews > Berlin > Mateo

Civil Status: In partnership/married # of children: 0 Highest level of education: Bachelor degree Employment: Self-Employed Age: 25 Location: Berlin

That's only Tuesdays and Thursdays if I remember correctly. And then I have, for the most part, the middle of the day free and depending on the day of the week it is, I have an afternoon class and it's also quite typical that I have evening classes but these are always changing and they're relatively flexible.

Mateo_Pos_3 Work situation > Flexibility (0)

It is very flexible. It's typically... my English sessions are planned. There are regular sessions but then there are also some very... if there's something else that's happening in the meantime, then it's quite flexible. It's possible to move things around.

Mateo_Pos_5 Work situation > Flexibility (50)

I would be happy... I would be happy to have a little bit of paid vacation but as far as changes to my daily routines... No, I think it works alright.

Mateo_Pos_11 Work situation > Workplace & Career (50)

I am pretty satisfied. It's, you know, I hadn't thought it in the

- Source:

<https://www.maxqda.com/document-analysis-software/>

Fully Automated Text Analysis

Advantages and Disadvantages

- Advantages of fully automated methods (Grimmer et al. 2022)
 - Maximum scalability: analyze millions of documents at near-zero marginal cost
 - Replicable: same input always produces the same output
 - No fatigue, no drift, no inter-coder disagreement (Lacy et al. 2015)
 - Enables longitudinal and comparative analyses across very large corpora
- Disadvantages (Grimmer et al. 2022)
 - Validity dependent on design choices and model specifications (dictionary, k , parameters, etc.)
 - Lack of human input: systematic errors are amplified
 - Thus, it still requires validation against human judgment

- Matches text with predefined word lists with words closely tied to the concept of interest (Young and Soroka 2012)
 - Output: relative frequency of words per category in each text
 - Common dictionaries: LSD (Young and Soroka 2012), PopDict (Gründl 2020), MFD (Hopp et al. 2020)
- Transparent and fast, requiring no labelled training data (Grimmer et al. 2022)

Dictionary Analysis - Simple Example

```
list_dict <- list(  
  economy = c("economy", "job", "employ", "gdp", "growth", "budget", "deficit", "debt"),  
  healthcare = c("health", "medicaid", "medicare", "insur", "hospit", "doctor", "patient"),  
  education = c("educ", "school", "teacher", "student", "colleg", "univers", "tuition"),  
  environment = c("climat", "environ", "energi", "green", "emiss", "renew", "pollut"),  
  immigration = c("immigr", "border", "citizen", "visa", "migrant", "deport"),  
  foreign_policy = c("foreign", "diplomat", "diplomaci", "allianc", "nato", "trade", "tariff", "sanction"),  
  social_security = c("retir", "pension", "senior", "elder", "social secur"),  
  crime = c("crime", "crimin", "polic", "justic", "prison", "law")  
)
```

- Exemplary dictionary for policy issue occurrence

Dictionary Analysis - Complex Example

Table 3: Proposed German populism dictionary

[illegible]

2004

The search terms include regular expressions:

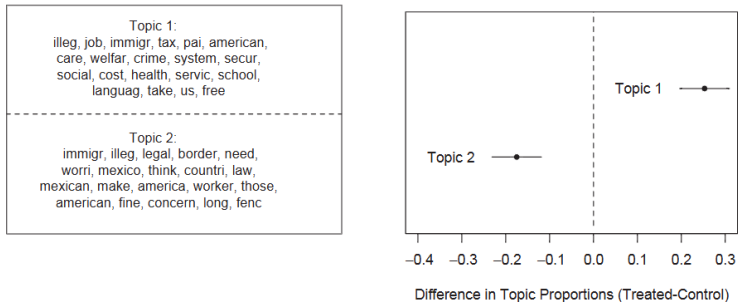
- German populist rhetoric dictionary by Gründl 2020

- Key limitations (Young and Soroka 2012)
 - Ignores context, syntax, and negation (“not happy” counts as “happy”)
 - Domain sensitivity: dictionaries built for one text type may fail in another
- Best practice: validate dictionary scores against a hand-coded gold standard (Grimmer et al. 2022)

- Scales political texts on a dimension based on differences in word frequencies (Slapin and Proksch 2008)
- Assumes more frequent use of specific words by each side of the spectrum
- Resulting dimension needs to be interpreted by the researcher
- Not guaranteed to be a classic left-right-dimension
 - Could be, e.g., issue salience, populist attitudes, government v. opposition
- Good for exploration of data

- Commonly done using Latent Dirichlet Allocation (LDA) (Blei et al. 2003)
- Assumption:
 - Documents are mixtures of topics
 - Topics are represented by specific words
- Fully unsupervised: no labels or prior knowledge of content required
- Output: topics described by most frequent words
 - Researcher interprets and labels topics
 - Key decisions: number of expected topics and pre-processing

FIGURE 7 Words and Treatment Effect Associated with Topic 1



- Topics in open-ended survey responses from a survey experiment by Roberts et al. 2014

Word Embeddings - 1

- Represent words as vectors in a continuous semantic space (Mikolov et al. 2013)
 - Words with similar meanings cluster together
- Word2Vec (Mikolov et al. 2013) and GloVe (Pennington et al. 2014)
 - Static embeddings
 - Each word has one fixed vector regardless of context
- Contextual embeddings (BERT, RoBERTa): word meaning varies by sentence (Devlin et al. 2018) → Outperforms static embeddings

- Common application:
 - Tracking semantic change over time (Hamilton et al. 2016)
 - Train embeddings separately on texts from different time periods
 - Measure shifts in meaning e.g., “right-wing”, “populism”, “culture”
 - Similarity Analysis (Lin 2025)
 - Uncover rhetorical similarity between political texts
 - E.g., to measure congruence of political parties

Semi-Automatic Text Analysis

Advantages and Disadvantages

- Advantages of semi-automatic methods (Grimmer et al. 2022)
 - Applicable to large datasets
 - Applicable to most classification tasks
 - Active learning further reduces labelling cost without sacrificing accuracy
 - Human judgement is involved in shaping what the model learns
- Disadvantages (Grimmer et al. 2022)
 - Training data can be costly to produce in sufficient mass
 - Risk of training for human biases
 - Model performance depends heavily on label quality
 - Danger of introducing overfitting of models on training data
 - Domain shift: E.g., models trained on news text may fail on social media, and vice versa (Ben-David et al. 2010)

- Semi-supervised extension of the basic dictionary approach to account for its caveats (Benoit and Laver 2003)
- Developed specifically for scaling texts on a policy dimension
- Requires per-coded texts → Wordscores attributes differences in word frequencies to the pre-labelled categories
- Texts are positioned relative to the pre-coded texts

Supervised Machine Learning

- Core idea: train on human-labelled data → apply the trained model on larger dataset (Grimmer et al. 2022)
 - Classic algorithms: Naïve Bayes, SVM, Logistic Regression, Random Forests
 - Advanced approaches: fine-tuned transformer models (BERT, RoBERTa) (Devlin et al. 2018)
- Workflow: label → train → validate → classify (Grimmer et al. 2022)
 - Validation against additional human-labelled data is mandatory
 - Common validity measures: precision, recall, F1

Structural Topic Models

- Extends LDA by allowing metadata to affect topic prevalence (Roberts et al. 2014)
 - e.g. “Does topic X appear more in texts marked as left-wing or right-wing press?”
- Same as plain topic models:
 - The algorithm discovers structure & humans interpret topics (Grimmer et al. 2022)
 - Researcher must choose the number of topics
- Advantage over plain topic models:
 - Covariates affect topic emergence and allow direct hypothesis testing
 - E.g., estimating the effect of party membership on topic usage in speeches or the words used to discuss the same topics

Large Language Models - 1

- LLMs used as zero- or few-shot coders for text classification tasks:
 - Can out-perform crowd coders on several tasks (Gilardi et al. 2023)
 - Can out-perform expert coders (Le Mens and Gallego 2025)
 - Can correlate highly with expert survey scores (Benoit, De Marchi, et al. 2025)
- No labelled training data required
- Highly flexible to new tasks and categories

Large Language Models - 2

- But: prompt engineering and validation require human effort
(Pangakis et al. 2023)
- Risks of using LLMs:
 - Small changes in wording can produce large changes in classifications
 - Model updates change outputs → reproducibility requires local model
 - Inaccessible systematic biases based on training data
- Novel approach, constantly in motion and changing & many publications still under review
- Using the API of proprietary models is not free and will be charged per token of input and output

B.2 Negative

By “negative”, we mean negative messaging, i.e., direct attacks, criticism, disparaging comments, or generally comments which expressly say or at least suggest that some person/event/outcome is bad/negative. Our understanding is distinct from other definitions that do not separate negative sentiment from merely negative events or outcomes. For example, while “Hurricane kills 50” is certainly bad, it is not negative messaging. In contrast, “Failure to address climate change means 50 have died in hurricane” is negative. In other words, a message is negative when it expresses negative sentiment directly towards an individual, group, or outcome and is not merely any message which contains negative valence.

C.2 Negative (no instructions)

create a table classifying the following messages as negative (1) or not negative (0):

- Prompt provided to LLM for sentiment analysis by Heseltine and Hohenberg 2024

Summary & Coding Session

- Three main strains of text analysis methods:
 - Manual Coding of Texts: Great validity, high costs for large datasets
 - Close Reading, Qualitative Content Analysis, Discourse Analysis
 - Fully Automated Analysis: Low cost for large data analysis, low validity without checks
 - Dictionary Analysis, Wordfish, Topic Models, Word Embeddings
 - Semi-Automatic Text Analysis: Compromise manual and fully automatic analysis
 - Wordscores, Supervised Machine Learning, Structural Topic Models, Large Language Models

- We will start coding now
- We will conduct the following steps:
 - Data pre-processing
 - Keywords analysis
 - Dictionary analysis
 - Sentiment analysis
 - Topic model analysis
 - Wordfish
 - Wordscores
- Please open the provided R file

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